Result 1 (assumed inclusion constructs certified data). [Topology foundations, certified inclusion constructor]
Let X be a type. Let S and T be subsets of X. Assume S is a subset of T (`inclusion`). Let $Q =$ certified inclusion$(X, S, T, \mathit{inclusion})$. Then

$$Q.\text{source} = S \text{ and } Q.\text{target} = T.$$

Result 2 (inclusion preserves the underlying element). [Topology foundations, Set.inclusion underlying value]
Let X be a type. Let S and T be subsets of X. Assume S is a subset of T (`inclusion`). Let x be an element of S. Let $f = \iota_{S,T}$. Then

$$f(x) = x.$$

Adapted from the cited source.

Result 3 (certified map preserves the underlying element). [Topology foundations, Set.inclusion underlying value]
Let X be a type. Let Q be a certified inclusion between subsets of X. Let x be an element of Q.source. Let $f = \iota_Q$. Then

$$f(x) = x.$$

Adapted from the cited source.

Result 4 (pair inclusion square commutes). [Topology foundations, commuting square of subtype inclusions]
Let X be a topological space. Let Q be a certified inclusion between pairs in X. Then

$$\text{ambient composite}(X, Q) = \text{relative composite}(X, Q).$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.